

Generalized Linear Model

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Family: binomial or Poisson. This specifies the variance function. This allows the generalized linear model to relax the assumption of constant variance of residuals.

Link functions: transform response variable (e.g. count, probability) by some functions (e.g. natural log, logit). The transformed response variable is called linear predictor η (predictor here means predicted value): $\eta = \beta_0 + \beta_1 x_1 + \dots$

η has a linear relationship with $\beta_0 + \beta_1 x_1 + \dots$:

$$\eta = \beta_0 + \beta_1 x_1 + \dots$$

Apply inverse link $f^{-1}(x)$ to η to transform it back to y :

$$f^{-1}(\eta) = f^{-1}(\beta_0 + \beta_1 x_1 + \dots)$$

$$y = f^{-1}(\beta_0 + \beta_1 x_1 + \dots)$$

The model will predict $\hat{y}$ by applying inverse link to $\beta_0 + \beta_1 x_1 + \dots$.

1 Link function: natural log (when y is count)

Link: $\eta = \ln(y)$

Inverse link: $y = e^\eta$

Example:

There is a generalized linear model where $\beta_0 = 1, \beta_1 = 1, x_1 = -5$:

$$\ln(y) = \eta = \beta_0 + \beta_1 x_1$$

The predicted value of $\ln(y)$ is $1 + 1 \times (-5) = -4$ (link scale). The predicted value of y is $e^{-4} \approx 0.018$ (data scale).

2 Link function: logit (when y is probability)

$$\text{odds} = \frac{p}{1-p}$$

$$\text{log-odds (logit)} = \ln\left(\frac{p}{1-p}\right)$$

$$\text{Link: } \eta = \ln\left(\frac{y}{1-y}\right)$$

$$\text{Inverse link: } y = \frac{1}{1+e^{-\eta}}$$

Example:

Calculate the probability p when log-odds is 0.

$$\ln\left(\frac{p}{1-p}\right) = 0$$

$$e^{\ln\left(\frac{p}{1-p}\right)} = e^0$$

$$\frac{p}{1-p} = 1$$

$$p = 1 - p$$

$$p = 0.5$$

Calculate the probability p when log-odds is 1.

$$\ln\left(\frac{p}{1-p}\right) = 1$$

$$e^{\ln\left(\frac{p}{1-p}\right)} = e^1$$

$$\frac{p}{1-p} = e$$

$$p = e - ep$$

$$(1+e)p = e$$

$$p = \frac{e}{1+e} \approx 0.73$$